

Subsystem Specification

Hull & structure

Provide a stable, unsinkable, modular catamaran platform that a 4-year-old can help assemble, carrying the electronics box, thruster pods, mast and DUPLO deck.

Document	BOATY-SSS-HUL
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	19 allocated SRS requirements → 25 subsystem requirements (19 M, 5 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Provide a stable, unsinkable, modular catamaran platform that a 4-year-old can help assemble, carrying the electronics box, thruster pods, mast and DUPLO deck.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Six hull segments with closed-cell foam cores • Two crossbeams with M3 rail • Electronics box, saddle and glands • Camera hood • DUPLO deck plate • Mast tube and socket • Carry handle • All structural fasteners	• Thruster pods (PRP) • Anything inside the box (PWR, HLM, MCP) • Flag, hoop, beacon, labels (REC)

1.3 Breakdown

Item	Description
HUL-1 Hull segments x6	Bow, mid and stern per hull; ≤ 200 mm long; foam-filled PETG shells
HUL-2 Crossbeams x2	Box-section beams with a 10 mm M3 hole grid on top
HUL-3 Electronics box	Clip-lock box, saddle, 4 × PG7 glands, key dock
HUL-4 Deck & mast	DUPLO plate, mast tube, socket, handle
HUL-5 Camera hood	Printed spray hood and window on the box front

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-16 Hull segment joint	internal	Owner	\$8
IF-17 Pod mount	Propulsion	Conforms	\$8
IF-18 Box & glands	Power	Owner	\$8
IF-19 Mast socket	Helm	Owner	\$8
IF-20 DUPLO deck	Operator & crew	Owner	\$8

4. Parent requirements

Primary (19): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
CAM-006	S	The lens shall be shielded from spray and cleanable on the bank without tools.	HUL-D19
REC-001	M	The boat shall stay afloat, carrying its maximum payload, with the electronics box flooded and any two hull segments...	HUL-D05, HUL-D06
MEC-001	M	The boat shall be a twin-hull catamaran no more than 650 mm long, 400 mm wide and 500 mm high (including mast).	HUL-D01
MEC-003	M	The boat shall carry at least 300 g of payload on the deck with at least 50 mm of freeboard.	HUL-D07
MEC-004	M	Static heel shall be 10° or less with 300 g at the deck edge. The boat shall not capsize in full-speed turns or in 100...	HUL-D04, HUL-D08, HUL-D09
MEC-005	M	Every printed part shall fit a 200 × 200 × 200 mm build volume (Ultimaker).	HUL-D02
MEC-006	M	Wetted and exposed printed parts shall be PETG or ASA. Hull segments shall be filled with closed-cell foam.	HUL-D05, HUL-D10
MEC-007	M	Hull segments, thruster pods, mast, electronics box and DUPLO deck shall be removable and refittable by hand, without...	HUL-D02, HUL-D14, HUL-D16, HUL-D25
MEC-009	M	The deck plate shall carry LEGO DUPLO-compatible studs. Genuine bricks shall grip firmly and be removable by a...	HUL-D17
MEC-012	M	The electronics box shall let in no water after 30 min at 0.3 m depth. Every cable entry shall go through a gland.	HUL-D15
MEC-014	S	The boat shall fit in a car boot and have a carry handle for one adult hand.	HUL-D21
MEC-015	S	Fasteners shall be stainless steel or plastic.	HUL-D12
ENV-003	M	The boat shall operate in waves up to 100 mm.	HUL-D09
ENV-004	S	The boat should keep operating in light rain. Heavy rain is outside the operating envelope.	HUL-D13
ENV-006	C	Exposed parts shall keep their function over two seasons of outdoor use.	HUL-D11
CHD-001	M	No child-handled part shall fit wholly inside the EN 71-1 small-parts cylinder.	HUL-D22
CHD-002	M	Child-handled parts shall have rounded edges (radius at least 0.5 mm) and no pinch points.	HUL-D23
CHD-003	M	The battery, electronics and propellers shall not be reachable without tools or an adult action.	HUL-D24
CHD-004	M	'Child tier' assembly (hull segments, pods, deck, flag) shall need no tools and be doable with small hands.	HUL-D14, HUL-D25

Supporting (5): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

CAM-005, MEC-008, MEC-013, PWR-010, ENV-002

5. Subsystem requirements

Requirements are numbered HUL-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Geometry and mass

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D01	Overall length shall be ≤ 620 mm and beam ≤ 360 mm. Height including the mast shall be ≤ 500 mm. Margins of 30 mm / 40 mm under MEC-001 for build tolerance.	M	I	BENCH	MEC-001
HUL-D02	Each hull shall comprise three segments (bow, mid, stern), each ≤ 200 mm long and printable in a $200 \times 200 \times 200$ mm volume.	M	I	BENCH	MEC-005, MEC-007
HUL-D03	Hull and structure mass (segments, beams, deck, box, mast, fasteners) shall be ≤ 1000 g.	M	T	BENCH	MEC-002
HUL-D04	Hull centreline spacing shall be ≥ 270 mm. Drives heel stability; see section 6.	M	A	BENCH	MEC-004

5.2 Buoyancy and stability

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D05	Closed-cell foam shall fill $\geq 90\%$ of each segment's internal volume.	M	I	BENCH	MEC-006, REC-001
HUL-D06	With the box flooded and any two segment shells breached, remaining buoyancy shall be $\geq 1.5 \times$ design mass (2.1 kg).	M	T	POOL	REC-001
HUL-D07	At design mass with 300 g on the deck, draft shall be ≤ 35 mm and freeboard ≥ 50 mm.	M	T	POOL	MEC-003
HUL-D08	Static heel with 300 g at the deck edge shall be $\leq 10^\circ$.	M	T	POOL	MEC-004
HUL-D09	The boat shall not capsize in full-speed (1.5 m/s) on-the-spot turns or in 100 mm waves with 300 g at the deck edge.	M	D	POOL	MEC-004, ENV-003

5.3 Materials and durability

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D10	Exposed and wetted printed parts shall be PETG or ASA, with ≥ 3 perimeters, ≥ 4 top/bottom layers and a hull shell ≥ 1.6 mm thick.	M	I	BENCH	MEC-006
HUL-D11	Exposed parts shall be UV-stable (ASA, or PETG with UV coating) and keep their function after two seasons.	C	I	LAKE	ENV-006
HUL-D12	Fasteners shall be A2/A4 stainless steel or plastic. No plain steel.	S	I	BENCH	MEC-015
HUL-D13	Deck and box lid shall shed rain: no water inside the box after a 10 min drip test, and no pooling on the deck.	S	T	BENCH	ENV-004

5.4 Modularity and interfaces

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D14	Segment joints shall conform to IF-16 (captive thumb-screws, dowels).	M	T	BENCH	IF-16, MEC-007, CHD-004
HUL-D15	The electronics box shall conform to IF-18 and pass 30 min at 0.3 m without ingress. Every cable entry uses a gland.	M	T	BENCH	IF-18, MEC-012
HUL-D16	The box saddle shall place the box lid ≥ 150 mm above the waterline and allow adult removal without tools.	M	I	BENCH	MEC-007, COM-002
HUL-D17	The DUPLO deck shall conform to IF-20.	M	T	BENCH	IF-20, MEC-009

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D18	The mast socket shall conform to IF-19, with the GNSS $\geq$ 150 mm from power wiring.	M	I	BENCH	IF-19, MEC-013
HUL-D19	The camera hood shall overhang the lens by $\geq$ 15 mm and leave the window reachable with a cloth, no tools.	S	I	POOL	CAM-006
HUL-D20	The camera mount shall hold the lens $\geq$ 120 mm above the waterline, level to $\pm 3^\circ$, facing forward.	S	I	BENCH	CAM-005

5.5 Handling and child safety

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D21	A carry handle within ± 50 mm of the centre of gravity shall carry 3 $\times$ design mass. With the mast removed, the boat shall fit a 650 $\times$ 400 $\times$ 250 mm envelope.	S	T	BENCH	MEC-014
HUL-D22	No child-handled part (segments, deck plate, thumb-screws, pod shells) shall fit wholly inside the EN 71-1 small-parts cylinder. Fasteners the crew handles are captive.	M	T	BENCH	CHD-001
HUL-D23	Child-handled edges shall have ≥ 0.6 mm chamfer or radius. Joints close face-to-face with no finger-trap geometry.	M	I	BENCH	CHD-002
HUL-D24	The box and battery bay shall open only by an adult action: latches needing ≥ 40 N or a two-handed release.	M	T	BENCH	CHD-003
HUL-D25	The crew's assembly steps (segments, pods, deck, flag) shall need no tools and ≤ 20 N finger force.	M	D	BENCH	CHD-004, MEC-007

6. Design analysis: buoyancy and stability

Hand estimates that size HUL-D04 to D09. They are checked against CAD volumes and then pool tests.

Quantity	Working	Result
Hull volume	$2 \times 600 \times 90 \times 100 \text{ mm} \times C_b 0.6$	6.5 L
Design mass	Boat 1.8 kg + payload 0.3 kg	2.1 kg
Draft	Displaced 2.1 L / waterplane 756 cm ² (Cwp 0.7)	$\approx 28 \text{ mm} (\leq 35 \text{ mm})$
Reserve buoyancy	6.5 L vs 2.1 L	3.1×
Two segments lost (worst case, no foam credit)	$6.5 \times 4/6$	$4.3 \text{ L} = 2.1 \times \text{design mass} (\geq 1.5 \times)$
Transverse BM (catamaran)	$2 \cdot A_{wp} \cdot (s/2)^2 / V$, $s = 270 \text{ mm}$	$\approx 0.66 \text{ m}$
Righting moment	$\Delta \cdot g \cdot GM \cdot \sin 1^\circ$, $GM \approx BM$	$\approx 0.24 \text{ N} \cdot \text{m per degree}$
Heel, 300 g at 180 mm	0.53 N·m / RM per degree	$\approx 2.2^\circ (\leq 10^\circ)$

A catamaran this wide is very stiff. The practical capsize risk is a wave or wake over a low bow, which is why HUL-D09 is demonstrated rather than calculated.

7. Design constraints and budgets

- ADD DD-14: printed foam-filled segments; PVC pontoons are the fallback
- Build volume 200 mm cube (Ultimaker)
- PETG default material; ASA for high-UV parts if needed

Budget	Allocation
Mass allocation	$\leq 1000 \text{ g}$ (estimate 930 g)
Cost allocation	£5 (box, glands); filament from stock
Power	None

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	8	1	9	1	-	19
POOL	3	1	1	-	-	5
LAKE	-	-	1	-	-	1

Ref	Open item
TBC-11	Final hull lines from CAD (IF-16)
TBC-13	Box selection and saddle (IF-18)
TBC-14	DUPLO stud size by test print (IF-20)